AI-DRIVEN SMART WORKER SAFETY AND EMERGENCY COORDINATION SYSTEM FOR PREDICTIVE HEALTH MONITORING AND INTELLIGENCE RESOURCE MANAGEMENT

ABSTRACT

Worker safety is an important concern in industries such as construction, manufacturing, mining, and warehousing, where workers may face physical and environmental hazards. Traditional monitoring systems mainly depend on manual inspections and fixed thresholds, which may not consider differences in individual workers' health conditions. This project proposes a personalized worker-safety risk prediction system using wearable sensors and an ESP32 microcontroller. The system is designed to monitor parameters such as heart rate, oxygen saturation (SpO_2), temperature, body movement, and environmental conditions. Personalized baseline normalization is used to compare current sensor readings with each worker's normal condition. Machine learning algorithms, including Logistic Regression, Random Forest, and XGBoost, are considered to classify risk levels as Low, Moderate, and High. SHAP is used to explain the factors influencing the predictions. An Android application is designed to display sensor readings, risk levels, and important contributing factors. Local alerts can warn workers when configured safety limits are exceeded. The proposed system aims to support continuous worker monitoring and improve safety awareness. Further testing is required to evaluate the system's accuracy, reliability, and effectiveness. .

Keywords : *Worker Safety, Wearable Sensors, Internet of Things (IoT), ESP32, Personalized Baseline Normalization, Machine Learning, Risk Prediction, Explainable Artificial Intelligence (XAI), SHAP, Android Application*

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